

Note 1.5 Nested Quantifiers

Introduction

This section we need to learn about the nested quantifiers and how to use nested quantifiers to express mathematical statements, English sentences and its negations.

Understanding Statements of Involving Nested Quantifiers

- **Definition:**

One quantifier is within the scope of another.

- **Thinking of Quantification as Loops:**

In working with quantifications of more than one variable, it is sometimes helpful to think in terms of nested loops. For example, to prove $\forall x \forall y P(x, y)$ is true, we can make an outer loop through all possible values for x and then make an inner loop for each x to judge whether each y satisfies $P(x, y)$.

The Order of Quantifiers

- **For all universal quantifications or all existential quantifications:**

$$\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$$

It is true when $P(x, y)$ is true for every pair x, y and false when there is a pair x, y for which $P(x, y)$ is false.

$$\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$$

It is true when there is a pair x, y for which $P(x, y)$ is true and false for every x, y $P(x, y)$ is false.

- **For Other Statement:**

The universal quantification and existential quantifications can not exchange since they are not equivalent.

Translating Mathematical Statement into Statements Involving Nested Quantifiers

- **Strategy:**

Find the relationship between the mathematical statement and logical expressions.

Translating from Nested Quantifiers into English

- **Strategy:**
 - Write out what the quantifiers and predicates in the expression mean.
 - Express this meaning in a simpler sentence.

Translating English Sentences into Logical Expressions

- **Strategy:**
 - Select suitable variables, proposition and connectors to represent the expressions in the English sentences.
 - Connect them together by logic.

Negating Nested Quantifiers

- **Strategy:**
Negate successively applying the rules for negating statements involving a single quantifier.
For example,

$$\begin{aligned}
 \neg \forall x \exists y (xy = 1) &\equiv \exists x \neg \exists y (xy = 1) \\
 &\equiv \exists x \forall y \neg (xy = 1) \\
 &\equiv \exists x \forall y (xy \neq 1)
 \end{aligned}$$